

Tactical DNA: Coaches, Styles, and Player Roles from Football Passing Networks

Project Progress Report for EECS 4414 Information Networks

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Abstract

Football tactics are debated qualitatively but rarely measured. We treat passing networks (directed weighted graphs whose edges count completed passes between players) as *tactical fingerprints* and use their structural signatures for three inference tasks: identifying a team’s coach from network features, discovering tactical archetypes by unsupervised clustering, and embedding players into a structural role space. Using StatsBomb open event data, we build per-match passing networks and a feature pipeline spanning centrality, global structure, and network motifs. This progress report presents the completed data-ingestion and feature pipeline, summary statistics characterizing the corpus, and initial coach-identification results on FC Barcelona across its successive coaching eras, a single-club setting that isolates the coach signal from the confound of team identity.

Keywords

network analysis, football analytics, passing networks, social network analysis, motif analysis, sports analytics

1 Introduction and Motivation

Football tactics are extensively debated in informal media but rarely quantified with rigor. Coaches, pundits, and fans speak of “possession football,” “gegenpressing,” or a team’s “style of play” as if these were measurable properties, yet such labels are typically applied qualitatively. Passing networks (directed weighted graphs whose nodes represent players and whose edge weights count completed passes from one player to another) provide a principled way to encode tactical structure as a measurable mathematical object [4, 9]. A growing body of work has shown that passing-network properties relate to team performance and offensive efficiency [2, 6], but most studies are *descriptive*: they characterize a single team or compare a small number of matches, without using network signatures as features for higher-level inference tasks.

We close this gap by treating passing networks as tactical fingerprints and using them to answer three concrete inferential questions, defined formally in Section 2: can we identify a team’s coach from the structure of its passing alone (*coach identification*); do matches cluster into recognizable tactical schools without supervision (*archetype discovery*); and can players be embedded into a structural role space that recovers their on-pitch position and surfaces structural “successors” to canonical playmakers (*player role embedding*)? The central motivating hypothesis is that a coach imposes a reproducible structural signature on a team’s passing, one strong enough to be recovered by a classifier and, potentially,

to persist when the coach changes clubs. The problem is important both as a test of whether qualitative tactical language has a measurable structural basis and for its applications in opponent scouting, recruitment (finding players who fill a structural role), and the automated summarization of playing style.

2 Problem Definition

Notation. Let $G_m = (V_m, E_m, w_m)$ denote the passing network of a team in match m , where V_m is the set of starting players, $E_m \subseteq V_m \times V_m$ is the set of directed pass relations, and $w_m : E_m \rightarrow \mathbb{Z}^+$ assigns each edge the number of completed passes from one player to another. To keep graphs comparable we fix $|V_m| = 11$ (the starting eleven; see Section 4.1). Let $\phi(G_m) \in \mathbb{R}^d$ be a feature embedding of G_m derived from its structural properties (centrality distributions, global metrics, and motif counts), and for Problem 3 let $\psi(p) \in \mathbb{R}^{d'}$ be an aggregated ego-network signature of player p across all matches in which they appeared.

Problem 1 (Coach Identification). Given a labeled set $\{(G_m, c_m)\}$ where $c_m \in C$ is the coach of the team in match m , learn a classifier $f : \mathbb{R}^d \rightarrow C$ minimizing the expected misclassification rate, equivalently maximizing accuracy and per-class macro-F1, such that $f(\phi(G_m))$ predicts c_m significantly above a naive baseline. A key sub-question is whether a coach’s signature persists across clubs. Most notably, we ask whether Pep Guardiola’s fingerprint is preserved from his Barcelona tenure (2008/09–2011/12) to his 2015/16 season at Bayern Munich, which tests directly whether the coach signal is club-bound or coach-bound.

Problem 2 (Tactical Archetype Discovery). Given an unlabeled set $\{G_m\}$, partition the matches into clusters $\{C_1, \dots, C_k\}$ that optimize an internal cluster-quality objective (silhouette, modularity). We then evaluate, *a posteriori*, whether the discovered clusters correspond to known tactical schools such as possession-based play, gegenpressing, or low-block counter-attack.

Problem 3 (Player Role Embedding). For each player p , learn a low-dimensional embedding $\Phi : \mathbb{R}^{d'} \rightarrow \mathbb{R}^2$ of the ego-network signature $\psi(p)$ that preserves structural similarity, evaluated by *position purity*: the fraction of each player’s k nearest neighbors in the embedding that share their on-pitch position. We complement this with qualitative case studies, identifying structural successors to canonical playmakers such as Xavi Hernández, Andrea Pirlo, and Luka Modrić.

2.1 Hardness

The three problems are hard along distinct axes. *Problem 1* is a statistical learning problem in a high-dimensional, small-sample,

and class-imbalanced regime (coaches with single-season tenures contribute few matches), where the principal difficulty is *identification*: distinguishing a genuine coach signal from the confounds of team identity, squad personnel, and opponent. Its feature-extraction subroutines also carry computational cost: betweenness centrality is computed in $O(|V||E|)$ time by Brandes’ algorithm, and exact motif (subgraph) counting is intractable in general, being closely tied to subgraph isomorphism (NP-complete) and #P-hard for counting; we sidestep this by fixing the motif size to three nodes, for which the triadic census is polynomial. *Problem 2* is hard in the optimization sense: minimizing the within-cluster sum of squares for k -means is NP-hard [1], and maximizing modularity is likewise NP-hard [3], so we necessarily rely on heuristics (Lloyd’s algorithm, agglomerative merging) with multiple restarts rather than exact optimization. *Problem 3* requires optimizing a non-convex embedding objective; neighbor-preserving methods such as UMAP have no global-optimality guarantee, and a naive all-pairs structural-similarity computation is $O(n^2)$ in the number of players. We discuss the practical mitigations for each in Sections 4 and 5.

3 Related Work

Passing networks as a model of tactics. López Peña and Touchette [9] introduced a network-theoretic treatment of football strategy, using centrality and flow measures on passing graphs to characterize team play; Buldú et al. [4] extended this with a multilayer, space- and time-aware analysis of passing dynamics. Both establish that passing networks encode tactically meaningful structure. Our work builds on their structural descriptors but differs in aim: rather than describing a single team or match, we use the descriptors as features $\phi(G_m)$ for supervised and unsupervised *inference* across hundreds of matches.

From description to inference. The systematic review of Alves et al. [2] catalogues social-network analysis in football and shows the literature is dominated by descriptive performance and tactical applications. This positions our contribution precisely: we treat the same network signatures as inputs to higher-level inference tasks (identification, clustering, embedding) that the descriptive body largely leaves unexplored.

Motifs. Milo et al. [8] established network motifs as the recurring building blocks of complex networks. Li et al. [6] applied triadic-motif analysis to football passing networks across eighteen competitions, finding that motifs with more bidirectional (back-and-forth) links are the most significant and that teams tend to recirculate the ball rather than play direct forward chains. This motivates our inclusion of motif frequencies as a feature family, whose discriminative value for Problems 1–2 we test empirically.

High-performance and temporal structure. Chacoma [5] identifies and optimizes high-performance passing networks and studies their temporal evolution within matches, motivating our planned temporal extension (per-segment networks).

Embedding methods. For the low-dimensional projections in Problems 2 and 3 we adopt UMAP [7], which preserves local neighborhood structure better than linear projections; we pair it with PCA for interpretability. All data are drawn from the StatsBomb open release [10].

4 Methodology

4.1 Data and Network Construction

We use the publicly available StatsBomb Open Data [10], which provides professional-grade event-level records, each annotated with a timestamp, pitch location on a standardized 120×80 grid, the players involved, and an outcome. For each match m we construct FC Barcelona’s passing network G_m over the *eleven-starter window*: all events preceding the first substitution by *either* team. This is the conservative choice: it guarantees that all twenty-two starters are on the pitch throughout, at the cost of ending the window early when the opponent substitutes first, which happens in more than half of matches (280 of 515 with a substitution) and trims some otherwise-valid Barcelona passing. Following established practice [4, 9], restricting to the eleven starters keeps $|V_m| = 11$ across all matches, removing the size confound that would otherwise distort centrality and motif statistics.

Starting elevens are read from the match lineups (players whose start reason is “Starting XI”) and cross-checked against the Starting XI tactics event.¹ An edge (u, v) is added for each completed open-play pass from player u to player v , weighted by the number of such passes; passes whose type is a throw-in, corner, free kick, goal kick, or kick-off are excluded so that edges reflect open-play structure, while passes immediately following a recovery or interception are retained as open play. We verified that the resulting corpus has no self-loops, all edge weights at least one, and exactly eleven nodes in every network.

4.2 Network Features

The feature embedding $\phi(G_m)$ aggregates three families of structural descriptors, each shown in prior work to carry tactical signal:

Centrality measures. For each node $v \in V_m$ we compute weighted in-degree and out-degree, betweenness centrality, eigenvector centrality, and PageRank. From the resulting per-node distributions we extract summary statistics (mean, variance, skewness, Gini coefficient), yielding a centralization profile that captures how hierarchically organized the team’s passing is [4].

Global structural metrics. Network density, weighted clustering coefficient, average shortest-path length, algebraic connectivity (the second-smallest eigenvalue of the graph Laplacian), and the largest eigenvalue of the adjacency matrix (spectral radius). Together these capture how cohesive a team’s passing organization is.

Network motifs. Motifs are small recurring subgraph patterns whose frequencies characterize the local structural “vocabulary” of a graph [8]. We enumerate the connected directed three-node motifs via the triadic census and record their normalized frequencies. As discussed in Section 2, fixing the motif size to three nodes keeps this computation polynomial [6].

For Problem 3, we construct a separate per-player feature vector $\psi(p) \in \mathbb{R}^{d'}$ from each player’s ego-network properties aggregated across all matches in which they appeared: ego degree, ego density, ego clustering coefficient, weighted in/out-degree, motif-position

¹One match (Granada vs. Barcelona, 2012/13) carries a lineups tagging error in which only two players are marked as starters; we recover the correct eleven from the intact Starting XI event.

frequencies (e.g., how often the player occupies the middle of a chain motif), and the spatial dispersion of pass endpoints.

Computational choices. Three definitions warrant note. For betweenness and weighted shortest-path computations we set the distance of an edge to the reciprocal of its pass count ($1/w$), since more passes denote a stronger and effectively shorter tie; using the raw weight as a distance would invert this relationship. Average shortest-path length is measured on the largest strongly connected component, as a handful of networks built from very short windows are not strongly connected. Eigenvector centrality is computed on the undirected weighted graph, whose symmetric adjacency matrix guarantees a real, well-defined leading eigenvector and which is the established convention in the passing-network literature [4, 9].

4.3 Algorithms

For **Problem 1**, we train three supervised classifiers on $\phi(G_m)$ with coach identity as the target: logistic regression as an interpretable linear baseline, a random forest to capture non-linear feature interactions, and XGBoost as a gradient-boosted ensemble. All are evaluated with stratified 5-fold cross-validation and class-weighted objectives to handle the imbalance arising from unequal match counts per coach.

For **Problem 2**, we apply k -means and hierarchical agglomerative clustering on $\phi(G_m)$, selecting k via the silhouette score and the elbow method, and visualize the result with UMAP [7] projections interpreted against known tactical schools.

For **Problem 3**, we project $\psi(p)$ into a two-dimensional role space using PCA (interpretable, linear) and UMAP (non-linear, neighborhood preserving), then use k -nearest-neighbor retrieval to identify structurally similar players.

All analysis is implemented in Python with NetworkX for graph construction and feature computation, scikit-learn for classical models and clustering, XGBoost for gradient boosting, and umap-learn for the projections. The full codebase and reproducible scripts will be released as a public GitHub repository.

4.4 Limitations and Difficulties

Three difficulties shape the approach. (i) Coach labels are not present in the event data and must be assembled from external records and aligned to match dates. (ii) Motif and centrality statistics are sensitive to graph size, which we neutralize by fixing $|V_m| = 11$. (iii) The richest within-club coaching variation in the open data lies with FC Barcelona; other clubs appear less densely (Section 5), which constrains the multi-team experiments to the final-report stage. (iv) A small number of networks (16 of 517 with fewer than 100 passes, 5 with fewer than 50) arise from very early substitutions and the resulting short windows; we retain them but flag them as low-sample.

5 Evaluation

5.1 Datasets

Our target corpus spans three competitive contexts: the La Liga “Big Three Golden Era,” Pep Guardiola’s 2015/16 Bundesliga season at Bayern Munich, and the men’s FIFA World Cups of 2018 and 2022. The StatsBomb open release supports all three: it provides every FC Barcelona La Liga match, both recent men’s World Cups, and Bayern

Table 1: Per-season composition of the corpus (FC Barcelona, La Liga 2005/06–2020/21). Mean passes and edges are per match over the eleven-starter window.

Season	Matches	Mean passes	Mean edges
2005/06	17	246.2	74.5
2006/07	26	273.9	74.5
2007/08	27	255.7	75.0
2008/09	31	256.8	73.4
2009/10	35	298.7	77.8
2010/11	33	351.9	80.4
2011/12	37	346.3	81.9
2012/13	32	344.3	77.7
2013/14	31	285.7	74.3
2014/15	38	366.6	83.8
2015/16	38	298.6	75.1
2016/17	34	300.9	77.7
2017/18	36	319.4	80.2
2018/19	34	323.1	79.8
2019/20	33	354.7	83.5
2020/21	35	338.1	81.7
Total	517		

Munich’s 2015/16 Bundesliga season, the last of which makes the planned Guardiola Barcelona→Bayern transfer experiment feasible on open data. For this progress report we focus on the portion with the richest within-club coaching variation, **FC Barcelona across its successive coaching eras**. We retain the sixteen La Liga seasons from 2005/06 to 2020/21 with adequate coverage (two earlier seasons with fewer than ten Barcelona matches were excluded as too thin for reliable per-season statistics), yielding **517 team-match passing networks**. Restricting to a single club isolates the coach signal from the confound of team identity, directly addressing the central question of Problem 1. Table 1 reports the per-season composition of the corpus. The broader multi-team corpus, adding Real Madrid and Atlético Madrid (which appear in the open La Liga data primarily as Barcelona’s opponents), Bayern Munich, and the World Cup national teams, will support the cross-club transfer experiment and the unsupervised clustering of Problems 1–2 in the final report.

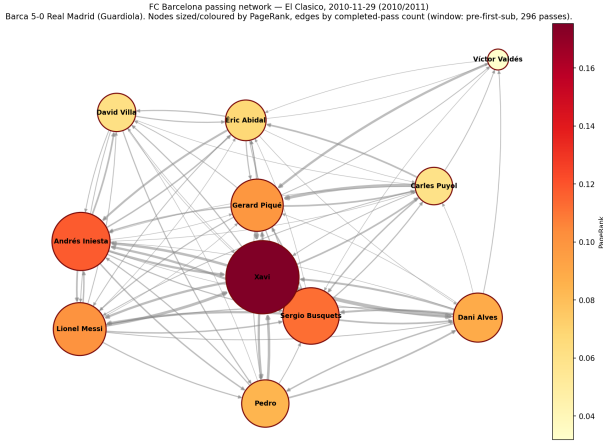
5.2 Summary Statistics of the Passing-Network Corpus

Across the 517 networks, Barcelona’s passing is dense and triangle-rich (mean density 0.71 ± 0.10 , mean weighted clustering 0.18 ± 0.04), consistent with the short-passing, possession-oriented style the side is known for. Per-match volume varies widely, from 18 to 660 completed open-play passes (median 309), because the window length depends on when the first substitution falls. Table 2 reports the global metrics over the full corpus.

The per-season breakdown reveals a measurable tactical signature. Cohesion metrics are lowest in the mid-2000s rebuilding seasons (density around 0.67–0.68 and algebraic connectivity around 11–13 in 2005/06–2008/09) and rise markedly through the trophy-laden 2010s. Individual metrics peak in standout seasons: weighted

Table 2: Summary statistics of the passing-network corpus (mean $\pm$ std across matches).

Metric	Mean $\pm$ Std
Number of edges	78.52 $\pm$ 11.17
Density	0.714 $\pm$ 0.102
Total completed passes	314.23 $\pm$ 110.58
Weighted clustering coefficient	0.181 $\pm$ 0.040
Avg. shortest-path length ($d = 1/w$)	0.400 $\pm$ 0.215
Avg. shortest-path length (hops)	1.296 $\pm$ 0.134
Algebraic connectivity (Fiedler)	16.21 $\pm$ 7.89
Spectral radius	32.92 $\pm$ 11.83

**Figure 1: FC Barcelona’s passing network in the 2010-11-29 El Clásico (5–0, Guardiola). Nodes are the eleven starters, sized and coloured by PageRank; directed edges are weighted by completed open-play pass count during the pre-substitution window. Xavi (centre) is the top-PageRank node, consistent with his role as the side’s passing metronome.**

clustering at 0.207 in Guardiola’s 2011/12, density at 0.761 in Luis Enrique’s 2014/15 treble side, and algebraic connectivity at 23.5 in 2019/20. That the networks become quantifiably more cohesive in the team’s strongest periods is an early signal supporting the project’s central hypothesis that tactical quality leaves a measurable structural footprint. Figure 1 shows a representative network: Barcelona’s 5–0 win over Real Madrid in November 2010, with Xavi the highest-PageRank node (the passing metronome), exactly the hierarchy the qualitative account would predict.

5.3 Evaluation Plan

Problem 1. We report accuracy and per-class macro-F1 under stratified 5-fold cross-validation and publish a confusion matrix exposing which coaches are confused with which. In the single-club setting the principal baseline is a majority-class classifier; in the final-report multi-team setting it becomes a team-identity classifier, so that a coach classifier outperforming it demonstrates the coach signal exceeds mere team identity. A second experiment trains on Guardiola’s Barcelona matches and tests on his Bayern Munich

season (or vice versa), measuring whether the signature transfers across clubs and leagues. A third ablates feature families (centrality, motifs, combined) to identify the most discriminative descriptors.

Problem 2. We report silhouette, modularity, and average intra- vs. inter-cluster distance across candidate k , then manually inspect each cluster against human-assigned tactical categories (possession, gegenpressing, low-block counter, direct). A UMAP projection annotated with landmark matches (marquee Clásicos from the 2010–2013 Pep-vs-Mourinho era and the 2022 World Cup Final) serves as the principal qualitative figure.

Problem 3. We test whether nearest-neighbor retrieval respects position (a midfielder’s neighbors should be predominantly midfielders), reporting position purity at $k = 5$ and $k = 10$ against a random-placement baseline, complemented by playmaker-successor case studies (Xavi, Pirlo, Modrić).

5.4 Preliminary Results: Coach Identification

We report initial Problem 1 results: an eight-class coach-identification task over the 517 FC Barcelona networks, with per-match labels assigned by date.² The class distribution is heavily skewed (Table 3): Guardiola contributes 136 matches while three coaches held the post for only a single season.

Against a majority-class baseline of 0.263, a class-weighted Random Forest reaches 0.358 ± 0.031 accuracy (macro-F1 0.229) under stratified 5-fold cross-validation, a +0.095 improvement; logistic regression trails at 0.244 (Table 4). An ablation (Table 5) shows that centrality features carry most of the discriminative signal, reaching 0.321 accuracy alone (nearly the full combined result), while motif frequencies are weak in isolation (0.242, barely above baseline). The confusion matrix (Figure 2) shows that Guardiola’s signature is by far the most distinctive (0.71 recall, 0.51 F1), whereas the three single-season coaches (Vilanova, Martino, Koeman) are essentially unidentifiable and collapse into the majority class.

Crucially, replacing random folds with a leakage-free *season-grouped* cross-validation (in which every season is kept intact within a single fold) drops Random Forest accuracy to 0.250 and macro-F1 to 0.113. Because each coach owns contiguous seasons with year-to-year roster turnover, random folds let same-season matches leak across train and test, so the model partly learns *which season* a match belongs to rather than *which coach*; the 0.108 accuracy gap quantifies this inflation. The honest reading is that network structure carries a measurable but modest coach signal above baseline, that much of the naive-CV gain is season-specific memorization, and that identifying a coach from a single season of matches is inherently hard. These findings directly motivate the final-report extensions: a larger multi-team corpus (Real Madrid, Atlético Madrid) so that every coach has support across rosters, the cross-club Guardiola→Bayern transfer experiment, and gradient-boosted models (XGBoost).

5.5 Risks and Trade-offs

Three risks remain. First, class imbalance in coach labels is significant: long-tenured coaches contribute far more matches than

²The 2019/20 season is split at Ernesto Valverde’s dismissal (13 January 2020) into his and Quique Setién’s spells, a more accurate ground truth than a season-level mapping. Xavi’s 2021/22–2023/24 tenure falls outside the free data, which ends at 2020/21.

Table 3: Coach label distribution (FC Barcelona, La Liga 2005/06–2020/21).

Coach	Matches	Seasons
Guardiola	136	2008/09–2011/12
Luis Enrique	110	2014/15–2016/17
Valverde	84	2017/18–2019/20 (pre-Jan)
Rijkaard	70	2005/06–2007/08
Koeman	35	2020/21
Vilanova	32	2012/13
Martino	31	2013/14
Setien	19	2019/20 (post-Jan)
Total	517	8 coaches

Table 4: Coach identification on FC Barcelona, under standard (random) and leakage-free (season-grouped) 5-fold cross-validation.

Model	CV	Accuracy	Macro-F1
Majority baseline	n/a	0.263	0.052
Logistic Reg.	random 5-fold	0.244 ± 0.012	0.215 ± 0.010
Random Forest	random 5-fold	0.358 ± 0.031	0.229 ± 0.048
Logistic Reg.	season-grouped	0.187 ± 0.057	0.104 ± 0.045
Random Forest	season-grouped	0.250 ± 0.090	0.113 ± 0.038

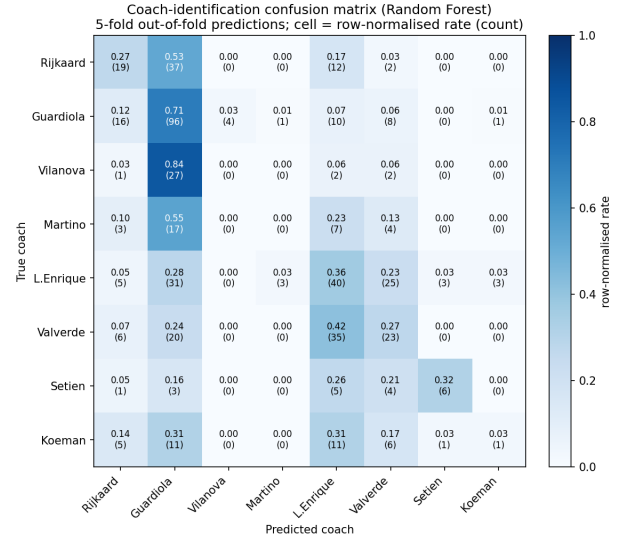
Table 5: Feature-family ablation (accuracy, stratified 5-fold CV).

Feature set	LR acc.	RF acc.
Centrality-only	0.205 ± 0.014	0.321 ± 0.022
Motifs-only	0.172 ± 0.050	0.242 ± 0.033
Combined	0.244 ± 0.012	0.358 ± 0.031

single-season ones; we mitigate with class-weighted objectives and stratified sampling. Second, motif counts are size-sensitive, controlled by fixing $|V_m| = 11$. Third, UMAP preserves local structure better than PCA but is harder to interpret semantically, so we report both. Time permitting, we will explore a temporal extension: per-segment networks within each match (e.g., 15-minute windows) that capture within-game tactical adaptation [5].

6 Conclusions and Future Work

This progress report establishes the foundation on which all three problems rest. We have implemented the full data-ingestion and passing-network construction pipeline over StatsBomb open data, computed a structural feature embedding spanning centrality, global structure, and motif families, and produced summary statistics characterizing the corpus (Section 5). For Problem 1 we report initial coach-identification results on FC Barcelona across its coaching eras: network features yield a measurable but modest coach signal (Random Forest accuracy 0.358 vs. a 0.263 baseline under standard cross-validation), carried chiefly by centrality, while a leakage-free

**Figure 2: Row-normalized confusion matrix for coach identification (Random Forest, out-of-fold predictions). Guardiola is the most reliably recovered; single-season coaches collapse into the majority class.**

season-grouped evaluation (0.250) shows that much of the naive gain is season-specific memorization and that single-season coaches are essentially unidentifiable.

Three directions remain for the final report. First, we will complete Problem 1 by extending to the multi-team corpus and running the Guardiola Barcelona→Bayern transfer experiment that tests whether the coach signal is club-bound or coach-bound. Second, we will carry out the unsupervised archetype discovery of Problem 2, mapping discovered clusters to known tactical schools. Third, we will construct the per-player ego-network embeddings of Problem 3 and evaluate position purity and playmaker-successor case studies. The temporal extension is a stretch goal if time allows.

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